Antonin Jaffré

+33 6 29 22 47 31 · antoninjaffre@gmail.com · LinkedIn · GitHub

Seeking a 6-month internship in financial markets and quantitative modelling, from March 2027

GitHub

Education

École des Mines de Saint-Étienne — Saint-Étienne, France
MSc in Applied Mathematics and Statistics · Mathematics in Action track (MAEA) — **2026 - 2027**

- National Master's degree **co-accredited with École Centrale de Lyon and ENS de Lyon**.
- *Mathematics and Machine Learning for Engineering* stream: probability and stochastic processes, Monte Carlo methods, optimisation, statistical learning, high-performance computing.

École des Mines d'Alès — Alès, France
Diplôme d'Ingénieur (5-year MSc-equivalent engineering degree) · AI & Data Science — **2024 - 2027**

- Double degree with École des Mines de Saint-Étienne. Machine learning, statistical modelling, software engineering.

Lycée Lafayette — Clermont-Ferrand, France
Intensive preparatory course (CPGE-ATS) for competitive entry to French engineering grandes écoles — **2023 - 2024**

Université Clermont Auvergne — Clermont-Ferrand, France
Two-year technical degree (DUT) in Physical Measurement and Instrumentation — **2021 - 2023**

Experience

Thales — Brive-la-Gaillarde, France
Internship: purchase price projection and prediction (Industrial Controlling) — **May - Aug. 2026**

- **Pricing engine (main deliverable).** Automated the annual year-ahead price projection campaign by translating management-control rules into a Python engine, with results delivered to buyers and automated reconciliation checks.
- **Machine learning price prediction.** Order-level price variation model built on ∼200,000 cleansed SAP purchasing records (28,000 part numbers, 2017–2026): LightGBM, Optuna, time-based cross-validation. **12.1% lower mean absolute error** than the incumbent method.

Phytocontrol Laboratory — Nîmes, France
Internship: economic performance analyst (Data & BI) — **Nov. - Dec. 2025**

- Pricing and profitability analysis (selling prices, cost of goods sold) on key accounts (Suez, Nestlé); financial data structuring in Power Query and Power BI dashboards for margin monitoring.

University of Wollongong — Wollongong, Australia
Research internship within an international team: nanoparticles for sunscreen UV filters — **2023**

Orange · Michelin — France
Internships: predictive maintenance, industrial analysis — **2023 - 2025**

Quantitative Projects & Leadership

S&P 500 cross-sectional equity scoring — Personal project · GitHub
Monthly stock selection, long-only and long-short, with full backtest — **2025 to present**

- 24 features built from raw prices (momentum, volatility, beta, drawdown, size, macro), ∼66,000 stock-months; L2 logistic regression coded from scratch (matches scikit-learn to 4 d.p.), *walk-forward* validation.
- Out-of-sample 2019–2024, net of costs: top-20% portfolio **26.0% p.a., Sharpe 0.93** vs. 0.85 for the S&P 500; CAPM shows excess return driven by beta (1.33), alpha +4.8% not significant ($t = 1.12$).

Machine Learning projects — École des Mines d'Alès
Algorithms from scratch & deep learning — **2024 - 2025**

- **Credit scoring:** decision tree implemented entirely without libraries. **Energy procurement optimisation:** neural network (TensorFlow/Keras) to optimise electricity purchase prices for municipalities in Occitanie.

Mines Alès Junior Étude — Alès, France
President, student-run consulting firm — **May 2025 to present**

- Led a team of 13; secured official CNJE accreditation; managed the project lifecycle and client partnerships.

Skills & Certifications

- **Technical:** Python (Pandas, NumPy, Scikit-learn, LightGBM, Optuna, TensorFlow), C, Java, SQL, R, Power BI, Excel (Power Query).
- **Certifications:** IBM Machine Learning · Yale Financial Markets (R. Shiller) · Stanford ML Specialization (A. Ng).
- **Languages:** French (native), English (TOEIC 885), Spanish (A2). **Interests:** quant investing, poker, swimming.